

Hill Cipher Encryption and Decryption

Encrypt ATTACK using the key CDDG. Find the ciphertext. Then, decrypt the ciphertext back to plaintext.

Encryption

Write the key and plaintext in matrix form. For the key CDDG, we have:

$$K = \begin{bmatrix} C & D \\ D & G \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

Then form the plaintext matrix:

$$P = \begin{bmatrix} A & T & C \\ T & A & K \end{bmatrix} = \begin{bmatrix} 0 & 19 & 2 \\ 19 & 0 & 10 \end{bmatrix}.$$

Note: By default, the Hill cipher uses a row-major key matrix and column-major plaintext vectors. Either way is acceptable, but the approach must be consistent.

Encryption is done as:

$$C = KP \mod 26 = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 & 19 & 2 \\ 19 & 0 & 10 \end{bmatrix} \mod 26 = \begin{bmatrix} 57 & 38 & 34 \\ 114 & 57 & 66 \end{bmatrix} \mod 26 = \begin{bmatrix} 5 & 12 & 8 \\ 10 & 5 & 14 \end{bmatrix}.$$

Therefore, the ciphertext is FKMFIO.

Decryption

Step 1: Find $\det(K)$

$$\det(K) = (2)(6) - (3)(3) = 12 - 9 = 3 \mod 26$$

Step 2: Find modular inverse of $\det(K)$

Since

$$\gcd(3, 26) = 1,$$

the modular inverse exists.

Solve for x in the linear diophantine equation:

$$3x + 26y = 1.$$

We find $x = 9$. A step-by-step approach can be found [here](#).

Step 3: Find K^{-1}

$$K^{-1} = 9 \times \text{adj}(K) \mod 26,$$

where

$$\text{adj}(K) = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix}.$$

Thus,

$$K^{-1} = 9 \times \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \mod 26 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}.$$

Step 4: Decrypt

Now,

$$P = K^{-1}C \mod 26 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 & 12 & 8 \\ 10 & 5 & 14 \end{bmatrix} \mod 26 = \begin{bmatrix} 0 & 19 & 2 \\ 19 & 0 & 10 \end{bmatrix}.$$

Which corresponds to the plaintext ATTACK.